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Learning Shared Latent Representations for Explainable Multimodal Breast Cancer Detection

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Abstract—Breast cancer stays one of the hardest problems in medical imaging. Mammography is the screening standard and it does show fine structural detail, yet sensitivity drops in dense tissue, and the ionizing dose is a known concern. Thermography, on the other hand, captures vascular and metabolic asymmetry with no radiation. Used alone it lacks specificity. This work builds a multimodal analysis pipeline. It fuses features from both imaging streams at the latent level. Mammograms flow through a pretrained ResNet50 backbone while Thermograms flow through MobileNetV2. Each embedding is projected into a shared 128 dimensional latent space and aligned using a mean squared error objective. Aligned representations are concatenated. Minimum Redundancy Maximum Relevance (mRMR) selection trims the fused vector before the classifier. A dense softmax head then produces the final call between the two clinical classes: benign and malignant. Training uses the CBIS-DDSM mammography dataset and the Visual DMR-IR thermography dataset. Splits are 70 percent train, 15 percent validation and 15 percent test, with class balance preserved across modalities. On the held out test set the proposed pipeline reaches 94.5 percent classification accuracy and a Precision Recall Area Under Curve of 0.991. Under Gaussian noise perturbation the accuracy only slips to 93.1 percent, which is a decent robustness margin. Latent space projections through t-distributed Stochastic Neighbor Embedding (t-SNE) and Principal Component Analysis (PCA) show clean separation between the benign and malignant clusters. SHAP based post hoc analysis then points at the small subset of latent dimensions doing most of the discriminative work. Overall, latent fusion looks like a promising path toward interpretable, research grade breast cancer analysis.

Index Terms—Breast cancer detection, thermography, mammography, multimodal fusion, latent space alignment, explainable AI, SHAP, CBIS-DDSM, DMR-IR

I. INTRODUCTION

Breast cancer sits among the leading causes of mortality among women worldwide. Over 2.3 million new cases are reported each year [1]. Mammography is still the current gold standard for screening [2], [3], but the diagnostic performance is bounded by several things at once: breast tissue density, exposure to ionizing radiation, and inter observer variability across radiologists. All this has driven a growing interest in add on imaging modalities [4], [5]. Infrared thermography stands out. It offers complementary physiological insight and needs no ionizing dose. It is also quick and painless to acquire.

Deep learning has quietly changed what is possible in both mammogram and thermogram analysis. Convolutional networks now pick out lesions, calcifications and vascular asymmetries that a radiologist can miss on visual inspection alone, especially at the earliest stage of disease where

the visible signal is small. Reported accuracies on standard mammographic corpora sit above 85 percent in recent years [6], [7], [8], and comparable numbers show up on thermal imaging [9], [10], [11]. Beyond raw accuracy, deep models can process thousands of screening cases in minutes. The clinical promise is early detection: catching malignancy patterns that hide inside pixel statistics but are too subtle for the human eye.

Thermography reads temperature asymmetries and vascular heat signatures. These reflect underlying metabolic and angiogenic changes that come with malignancy [12], [13]. Recent progress in machine learning and deep neural networks has revitalized thermographic imaging by allowing automated feature extraction from complex thermal patterns [9], [14]. Thermography on its own is often not specific enough though. It struggles when the task is to separate benign from malignant heat anomalies [15].

Mammography, in contrast, gives high resolution anatomical and structural detail. It picks up microcalcifications and soft tissue densities. Thermography simply cannot visualize these. That said, mammographic interpretation suffers in dense breasts. This leads to false negatives and reduced sensitivity in younger populations [16], [17]. These complementary strengths and weaknesses are exactly what motivates multimodal fusion of thermography and mammography [18], [19].

Clinical adoption of any such model needs one more thing. It needs explainability. Radiologists will not trust a black box that emits a score with no reasoning attached. Post hoc interpretability tools such as SHAP have become important precisely because they identify which features (or which regions of a scan) drove a given prediction. This is a prerequisite both for informed clinical disagreement with the model and for the kind of regulatory audit trails that any real deployment demands.

The present study develops a multimodal analysis framework that pulls features from thermography and mammography for early breast cancer detection. The thermographic branch uses the Visual DMR-IR dataset [20], [21], with quadrant wise segmentation and temperature normalization to capture localized vascular and metabolic patterns. The mammographic branch uses the CBIS-DDSM dataset [22]. Structural features are extracted from it through deep convolutional encoders. Both modality streams are then mapped into a shared latent representation, and their joint discriminative behaviour is examined. The point is to see how structural and physiological imaging complement each other for breast

abnormality classification.

The emphasis, to be clear, is not on shipping a complete clinical CAD system. It is on understanding the synergistic diagnostic value of combining these modalities inside the specific context of thermal biology. The findings suggest that thermography does carry biologically relevant temperature signatures. Those signatures complement structural mammographic features once they are integrated through latent level multimodal fusion.

II. LITERATURE REVIEW

Recent advances in breast cancer imaging cluster around three lines: mammography, thermography and multimodal fusion. The 2024 to 2025 cycle in particular has pushed validation across diverse datasets and complementary imaging modalities.

On the mammography side, Shen et al. [6] trained a deep network end to end on CBIS-DDSM. The task was binary benign versus malignant classification. They reported around 86 percent accuracy, one of the earliest reliable numbers on this corpus. Ragab et al. [7] combined a CNN feature extractor with a Support Vector Machine head, again on CBIS-DDSM. They reported around 87 percent for the same binary label set (benign versus malignant). Al-Mnayyis et al. [8] developed a CNN classifier for mammographic cancer detection using the KAUH-BCMD dataset. It contains 7,205 images from over 5,000 patients. They reported consistent accuracy around 96 percent for the binary benign versus malignant classes across heterogeneous data. Their message was blunt: large, clinically diverse validation sets matter. Chen et al. [23] extended this line by integrating mammography with ultrasound for lesion classification (binary: benign or malignant), reporting around 91.5 percent test accuracy.

In the thermography domain, Jalloul et al. [24] compared classical and deep learning approaches for infrared breast imaging. They reported classification accuracies of 94 to 98 percent using optimized feature extraction and Support Vector Machines. Their label set was binary (benign versus malignant). Their result confirmed that thermal imaging encodes diagnostically useful physiological information. Alzahrani et al. [11] built on that. They designed an end to end CNN enhanced with particle swarm optimization to classify malignant versus benign thermograms. They reached nearly 98 percent accuracy. This further demonstrated the viability of thermal signatures once learning architectures are optimized well. Roslidar et al. [10] surveyed deep learning approaches applied to DMR-IR thermograms and reported a wide accuracy range: 85 to 97 percent, with binary settings (benign versus malignant) consistently easier than three class settings (healthy, benign, malignant). Taken together these studies argue that deep learning is a good fit for thermographic breast analysis, even though the datasets remain small.

Parallel to the unimodal side, several studies have chased multimodal fusion so as to combine structural and physiological imaging. Wei et al. [25] proposed a multimodal framework that aligned ultrasound views with radiology text reports through dual encoders and cross modal feature alignment,

reporting around 90 percent binary classification accuracy. Li et al. [26] provided a systematic review of fusion methodologies. They summarized feature level, decision level and hybrid multimodal fusion strategies. Hou [27] carried out a bibliometric analysis of AI assisted multimodal breast imaging and reported a sharp research shift toward integrating MRI, mammography and thermography. Mostafa and Hassanien [18] experimented specifically with feature level fusion of thermographic and mammographic streams and reported around 92 percent accuracy for the binary benign versus malignant setting. Khan et al. [19] took a hybrid deep learning route with thermography plus mammography and reported around 94 percent binary accuracy on a mixed cohort.

Collectively, the literature indicates that recent mammography studies have pushed classification performance through larger datasets and cross center validation. Thermography deep models, meanwhile, have demonstrated the diagnostic relevance of physiological heat patterns quite clearly. That said, many existing studies stay unimodal, or skip an interpretable multimodal alignment mechanism, or rely on restricted dataset diversity. The trend toward validated multimodal designs mirrors the motivation of the present work. This work integrates thermal and mammographic modalities through latent level multimodal fusion for interpretable breast cancer analysis.

III. METHODOLOGY

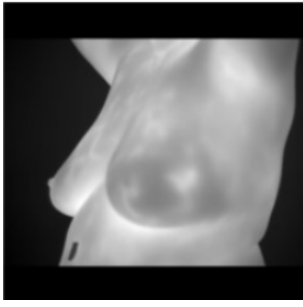
The proposed multimodal system integrates mammographic and thermographic imaging for robust breast cancer analysis. The design implements feature extraction pipelines that employ deep convolutional backbones, ResNet50 for mammograms and MobileNetV2 for thermograms, followed by standardized fusion, reduction and explainability layers. The implemented code based workflow has three critical stages: (i) modality specific preprocessing and normalization, (ii) deep feature extraction and encoding, and (iii) multimodal feature fusion and classification.

A. Dataset Description

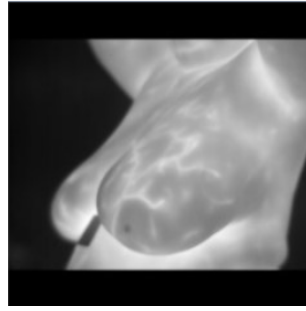
1) *Mammography Dataset: CBIS-DDSM*: The mammographic data are derived from the Curated Breast Imaging Subset of the Digital Database for Screening Mammography (CBIS-DDSM) [22]. This dataset is an updated and standardized version of the original DDSM repository, curated by the Cancer Imaging Archive (TCIA). It contains 2,620 ROI images extracted from digitized film mammograms encompassing approximately 1,500 patients. Each image is annotated with:

- Lesion type: *mass, calcification*
- Pathology: *benign, malignant*
- Breast density (BI-RADS I to IV)
- Bounding boxes and pixel level lesion masks

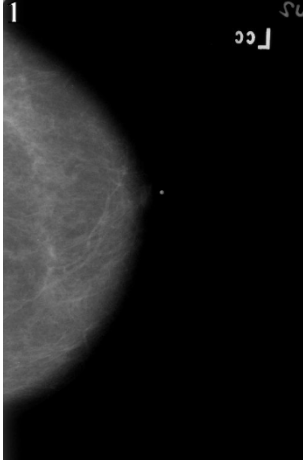
All mammograms were converted from 16 bit grayscale to 8 bit, normalized to the range $[0, 1]$, and resized to 224×224 pixels. The images were then reshaped into 3D tensors of size $224 \times 224 \times 3$ by channel replication so that they became compatible with convolutional backbones.



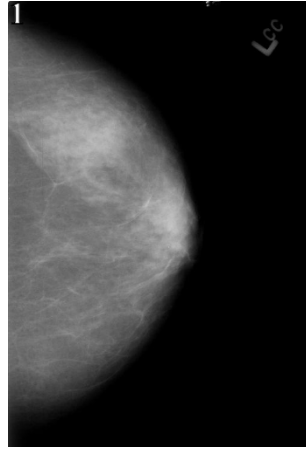
(a) Benign thermographic sample.



(b) Malignant thermographic sample.



(c) Benign mammographic sample.



(d) Malignant mammographic sample.

Figure 1: Representative thermographic and mammographic samples used in this study. Thermographic images illustrate temperature distribution differences, while mammographic images demonstrate structural variations between benign and malignant cases.

2) *Thermography Dataset: Visual DMR-IR*: The thermographic data were obtained from the Visual DMR-IR dataset [20], [21]. It consists of 200 participants imaged under standardized ambient conditions (20 to 22 degrees Celsius, 40 to 60 percent humidity). Each subject was imaged using a FLIR A320 infrared camera capturing frontal and oblique views of both breasts. Each thermogram is a temperature matrix $T(x, y)$ corresponding to surface thermal emission in Kelvin. Each is categorized as *healthy*, *benign*, or *malignant* based on biopsy confirmed outcomes. Representative samples from both imaging modalities are shown in Fig. 1. The thermographic images illustrate differences in thermal distribution patterns, while the mammographic images demonstrate structural variations associated with benign and malignant breast abnormalities.

a) *Sampling rationale*: A note on data volume is in order. Both datasets ship with more raw material than we actually use here. CBIS-DDSM offers thousands of full mammograms and their ROI crops; we sampled a balanced subset of 2,620 ROI images so as to match approximately what the DMR-IR side could support, and so as to keep the two class distributions comparable across modalities. On the thermography

side, DMR-IR has been recorded from 200 subjects, and we sampled frontal and oblique frames from a subset of these to keep acquisition geometry consistent. The choice to sample, rather than run every available image, was driven by two things. First, we wanted class balance across the two datasets so that the latent alignment step would not learn a spurious modality prior. Second, using the full corpora at their native size would have shifted the emphasis away from the fusion mechanism itself, which is the actual object of study here.

To enable multimodal analysis, mammographic and thermographic samples were aligned at the class level based on diagnostic category labels rather than direct patient wise correspondence, since the CBIS-DDSM and Visual DMR-IR datasets come from independent cohorts. The datasets were partitioned independently into training, validation and testing splits using a 70 / 15 / 15 percent ratio while preserving class balance across modalities. For the binary classification setup, DMR-IR healthy and benign cases were grouped into the benign class, so that both modalities carry the same two label semantics: benign and malignant.

B. Preprocessing Pipeline

Fig. 2 illustrates the complete workflow of the proposed multimodal thermo mammographic framework, including preprocessing, feature extraction, latent alignment, fusion, classification and explainability stages. The preprocessing phase ensures modality specific radiometric consistency and prepares both image streams for deep feature embedding. For a given mammogram $I_m(x, y)$ and thermogram $I_t(x, y)$, the overall transformation is formalized as:

$$\mathcal{P} = \{P_M(I_m), P_T(I_t)\} \rightarrow \{I'_m, I'_t\} \quad (1)$$

where P_M and P_T denote the preprocessing operators applied to mammography and thermography respectively.

1) *Mammogram Preprocessing (ResNet50 Pipeline)*: The mammographic preprocessing routine begins by reading raw DICOM or digital mammogram images and converting them into 8 bit grayscale arrays. The following steps are then executed in order:

- 1) **Histogram Equalization**: The grayscale image is contrast normalized using adaptive histogram equalization to enhance lesion boundaries and glandular tissue visibility [28].
- 2) **Standardization**: Pixel intensities are normalized to zero mean and unit variance so that acquisition bias is removed:

$$I'_m(x, y) = \frac{I_m(x, y) - \mu_m}{\sigma_m} \quad (2)$$

where μ_m and σ_m denote the mean and standard deviation of pixel intensities respectively.

- 3) **Dynamic Range Scaling**: Normalized values are linearly rescaled to $[0, 255]$ so as to preserve the image's dynamic range:

$$I''_m(x, y) = 255 \times \frac{I'_m(x, y) - \min(I'_m)}{\max(I'_m) - \min(I'_m)} \quad (3)$$

- 4) **Resizing and RGB Expansion**: The mammograms are resized to 224×224 and replicated across three channels

to form $I_m''' \in \mathbb{R}^{224 \times 224 \times 3}$. This step ensures compatibility with the CNN architectures used downstream [29].

The processed image is then passed to the ResNet50 backbone for deep feature embedding. This leverages pretrained convolutional kernels that extract hierarchical structural patterns.

2) *Thermogram Preprocessing (MobileNetV2 Pipeline):* Thermal imaging needs specialized handling because of varying emissivity and thermal contrast. The preprocessing here ensures consistent temperature scaling and noise suppression prior to deep encoding.

- 1) **Input Normalization:** Raw thermal maps are converted from absolute temperature matrices to 8 bit grayscale intensity maps. Normalization is performed as:

$$T'(x, y) = \frac{T(x, y) - \min(T)}{\max(T) - \min(T)} \times 255 \quad (4)$$

This transformation ensures uniform pixel intensity scaling across subjects.

- 2) **Noise Suppression:** Gaussian blur with kernel size (3×3) removes sensor noise and thermal inconsistencies without degrading vascular features.
- 3) **Segmentation:** A binary mask is generated so as to isolate the breast region based on intensity thresholding:

$$M_t(x, y) = \begin{cases} 1, & \text{if } T'(x, y) > \theta \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

where θ denotes a fixed temperature based threshold.

- 4) **Resizing and Channel Duplication:** Each thermogram is resized to 224×224 pixels and converted to a three channel format so that it becomes compatible with the deep network downstream [30].

Both preprocessing pipelines ensure that mammographic and thermographic tensors are normalized, spatially consistent and radiometrically calibrated for deep learning based feature extraction.

To improve generalization and reduce overfitting, data augmentation was applied during training. Mammographic and thermographic images were augmented using random horizontal flipping, rotational transformations within $\pm 10^\circ$, minor zoom variations and small intensity perturbations. These augmentations improve robustness to acquisition variability. They also enhance the ability of the model to learn invariant thermal and structural representations.

C. Feature Extraction Framework

Feature extraction employs convolutional backbones pretrained on ImageNet, then fine tuned so as to capture domain specific representations. Mammography and thermography branches operate independently before multimodal fusion.

a) *Why ResNet50 for mammography and MobileNetV2 for thermography:* The two backbones were not chosen at random. Mammograms carry rich high frequency structural information: microcalcification clusters, spiculations, tissue density gradients. Residual connections in ResNet50 let very deep filter stacks propagate gradients cleanly, which is exactly what one wants for these fine grained structural cues. The

2048 dimensional embedding it produces is expressive enough to encode subtle structural anomalies without collapsing them into a shorter code. Thermograms, on the other hand, are lower in spatial detail, noisier at the sensor level, and the DMR-IR corpus is much smaller than a typical natural image dataset. Deep heavy backbones tend to overfit here. MobileNetV2, with its inverted residuals and depthwise separable convolutions, is a much better fit in this regime. Its compact 1280 dimensional embedding is well suited to smaller data volumes and captures the coarser vascular and metabolic asymmetries that the thermal signal actually carries. In short: ResNet50 gives us depth where depth pays off; MobileNetV2 gives us compactness where compactness is a defence against overfitting.

1) *Mammographic Feature Encoding:* Each preprocessed mammogram I_m''' is fed into the ResNet50 convolutional backbone [29]. The final fully connected layer is removed, and global average pooling is applied so as to yield a compact, high level feature representation:

$$f_M = \phi_M(I_m'''; \theta_M) \quad (6)$$

where ϕ_M represents the feature extraction function and θ_M are the pretrained parameters. The resulting feature vector $f_M \in \mathbb{R}^{2048}$ captures complex structural and textural information such as tissue density, microcalcifications and spiculations that matter for malignancy prediction.

2) *Thermographic Feature Encoding:* Thermographic data are processed using MobileNetV2, a lightweight network optimized for computational efficiency and reduced overfitting on comparatively smaller datasets [30]. Each normalized thermogram I_t' is passed through MobileNetV2 as:

$$f_T = \phi_T(I_t'; \theta_T) \quad (7)$$

where ϕ_T represents the thermographic encoder and θ_T are pretrained weights. The extracted features $f_T \in \mathbb{R}^{1280}$ capture vascular, metabolic and symmetry related patterns that correspond to potential malignancy indicators.

D. Multimodal Feature Fusion and Classification

Once modality specific feature vectors f_M and f_T are in hand, they are concatenated at the feature level so as to form a unified multimodal representation:

$$\mathbf{F}_{fusion} = [f_M || f_T] \quad (8)$$

This fusion captures complementary information, combining the structural sharpness of mammography with the functional insight of thermography [?].

Concatenated deep representations tend to introduce feature redundancy and correlated latent activations though, especially in multimodal settings. So dimensionality reduction is performed using the Minimum Redundancy Maximum Relevance (mRMR) criterion [31]:

$$S^* = \arg \max_S \left[\frac{1}{|S|} \sum_{x_i \in S} I(x_i; c) - \frac{1}{|S|^2} \sum_{x_i, x_j \in S} I(x_i; x_j) \right] \quad (9)$$

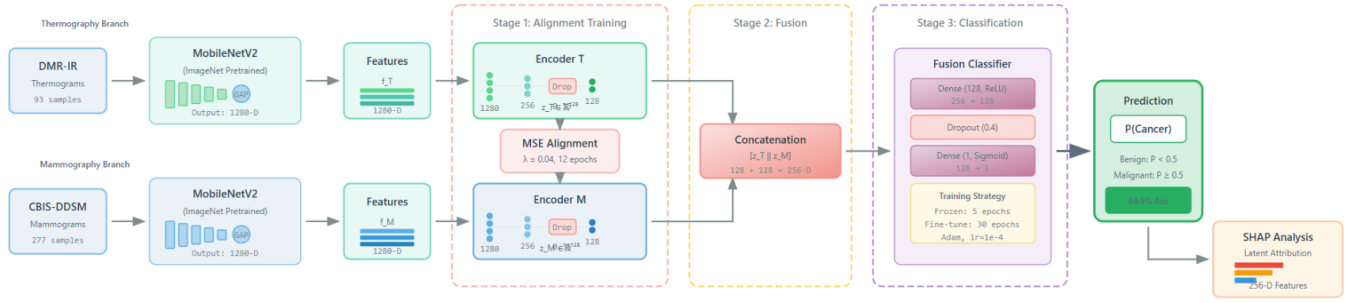


Figure 2: Overall workflow of the proposed multimodal thermo mammographic breast cancer detection framework, including preprocessing, feature extraction, latent space alignment, multimodal fusion, classification and SHAP based explainability.

where $I(\cdot)$ denotes mutual information and c is the class label. The reduced feature subset $\mathcal{S}^*$ is then fed into a fully connected neural classifier:

$$\hat{y} = \text{Softmax}(W_c \mathbf{F}_{\text{fusion}} + b_c) \quad (10)$$

The classifier predicts the probability of malignancy using categorical cross entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^C y_{ik} \log(\hat{y}_{ik}) \quad (11)$$

where y_{ik} and $\hat{y}_{ik}$ represent ground truth and predicted probabilities for class k respectively.

E. Explainability with SHAP

Post hoc interpretability is achieved using SHAP (SHapley Additive exPlanations) [32]. SHAP provides a unified measure of feature importance by decomposing the model output into additive contributions of each input variable:

$$f(\mathbf{x}) = f(\mathbf{x}') + \sum_{i=1}^n \phi_i \quad (12)$$

where ϕ_i represents the marginal contribution of feature x_i to the prediction. SHAP values were computed for both modalities and mapped so as to visualize how individual features (for example lesion asymmetry, temperature gradient) influenced the malignancy prediction. This makes the decision process transparent, which is important for interpretable medical decision support systems.

Algorithm 1 Multimodal Feature Extraction and Fusion for Breast Cancer Classification

Require: $\mathcal{D}_M = \{I_m^{(i)}, y^{(i)}\}_{i=1}^N$, $\mathcal{D}_T = \{I_t^{(i)}, y^{(i)}\}_{i=1}^N$
Ensure: $\hat{y}^{(i)} \in \{\text{benign}, \text{malignant}\}$ for each subject i

- 1: Initialize parameters $\theta_M, \theta_T, W_c, b_c$
- 2: Load mammographic and thermographic datasets; perform independent 70 / 15 / 15 percent train, validation and test splits while preserving class balance

▷ — *Preprocessing* —

- 3: **for** each subject $i = 1$ to N **do**
- 4: Histogram equalize $I_m^{(i)}$; normalize via Eq. (2); rescale via Eq. (3)
- 5: Resize and replicate channels $\Rightarrow I_m''' \in \mathbb{R}^{224 \times 224 \times 3}$
- 6: Normalize $T^{(i)}$ via Eq. (4); apply Gaussian blur; threshold mask via Eq. (5)
- 7: Resize thermogram $\Rightarrow I_t' \in \mathbb{R}^{224 \times 224 \times 3}$
- 8: **end for**

▷ — *Feature Extraction* —

- 9: **for** each subject $i = 1$ to N **do**
- 10: $f_M^{(i)} \leftarrow \phi_M(I_m'''; \theta_M)$ ▷ ResNet50, $f_M \in \mathbb{R}^{2048}$
- 11: $f_T^{(i)} \leftarrow \phi_T(I_t'; \theta_T)$ ▷ MobileNetV2, $f_T \in \mathbb{R}^{1280}$
- 12: **end for**

▷ — *Fusion and Dimensionality Reduction* —

- 13: **for** each subject $i = 1$ to N **do**
- 14: $\mathbf{F}_{\text{fusion}}^{(i)} \leftarrow [f_M^{(i)} \| f_T^{(i)}]$ ▷ Eq. (8)
- 15: $\mathcal{S}^* \leftarrow \text{mRMR}(\mathbf{F}_{\text{fusion}})$ ▷ Eq. (9)
- 16: **end for**

▷ — *Training* —

- 17: **for** each mini batch $\mathcal{B} \subseteq \mathcal{S}^*$ **do**
- 18: $\hat{y} \leftarrow \text{Softmax}(W_c \mathbf{F}_{\text{fusion}} + b_c)$ ▷ Eq. (10)
- 19: Compute $\mathcal{L}$ via Eq. (11)
- 20: Update $\{\theta_M, \theta_T, W_c\}$ via Adam, $\eta = 10^{-4}$
- 21: **end for**

▷ — *Evaluation and Explainability* —

- 22: Evaluate on test set: Accuracy, Precision, Recall, AUC
- 23: Compute SHAP values ϕ_i via Eq. (12)
- 24: Visualize SHAP heatmaps and return $\{\hat{y}^{(i)}, \phi_i\}$

F. Experimental Configuration

All experiments were carried out using Python and TensorFlow on an NVIDIA GPU enabled environment. The multimodal framework was trained using the Adam optimizer with a learning rate of 10^{-4} and categorical cross entropy loss. Training ran for 50 epochs with a batch size of 32 under mini batch optimization. Early stopping was applied so as to reduce overfitting. Performance evaluation was carried out using accuracy and PR AUC metrics on held out test data. Gaussian noise perturbation experiments were additionally performed to evaluate robustness under input variability.

G. Implementation Details

The proposed framework was implemented using TensorFlow and Keras in Python. ResNet50 and MobileNetV2 backbones pretrained on ImageNet were fine tuned for mammographic and thermographic feature extraction respectively. During preprocessing, all input images were resized to 224×224 pixels and normalized prior to feature encoding.

The ResNet50 encoder generated 2048 dimensional mammographic embeddings, while MobileNetV2 produced 1280 dimensional thermographic embeddings. These representations were projected into a shared 128 dimensional latent space using fully connected embedding layers with ReLU activation. Latent alignment was optimized using mean squared error loss during pretraining.

The fused latent representation was then passed through a lightweight fully connected classifier consisting of dense and dropout layers before softmax prediction. Model training was performed with the Adam optimizer at a learning rate of 10^{-4} and batch size of 32. Early stopping and validation monitoring were used so as to reduce overfitting during training.

IV. RESULTS AND ANALYSIS

A. Ablation Study and Comparative Analysis

To quantify the contribution of individual components of the proposed framework, an ablation study was carried out. It systematically evaluated unimodal baselines, multimodal fusion without latent alignment, and the full proposed model. As shown in Table I, multimodal fusion consistently outperforms unimodal approaches. This confirms the complementary nature of thermographic and mammographic information for the binary benign versus malignant task. The mammography only branch reaches 89.4 percent accuracy and PR AUC 0.941. The thermography only branch trails at 86.2 percent accuracy and PR AUC 0.912. Naive concatenation without alignment pushes accuracy to 91.8 percent. Adding the latent alignment step lifts it further to 94.5 percent, with PR AUC 0.991.

Incorporating latent space alignment yields a clear performance gain over unaligned fusion. This confirms its role in reducing modality induced representation discrepancies. The robustness evaluation under noisy conditions indicates only a marginal performance degradation, which highlights the stability and generalization capacity of the proposed approach.

Table II places the proposed model against past work on the same CBIS-DDSM and DMR-IR benchmarks. On

CBIS-DDSM alone, the strongest unimodal mammography baselines cluster around 86 to 87 percent accuracy [6], [7]. On DMR-IR alone, thermography deep models reach anywhere from 85 to 98 percent depending on architecture and label granularity [24], [11], [10]. Existing multimodal designs that combine mammographic and thermographic streams sit at 92 to 94 percent [18], [19]. The proposed latent fusion pipeline reaches 94.5 percent binary accuracy, which is on par with or above these multimodal baselines and considerably above unimodal mammography. The comparison is indicative and not head to head, since cohort composition and class definitions vary between studies. Still, the direction is consistent: adding thermography to mammography, and aligning the two representations, helps.

B. Multimodal Latent Space Alignment

To integrate heterogeneous imaging modalities effectively, separate encoders were trained for thermographic and mammographic feature representations. Each encoder compressed modality specific feature vectors of dimension 2048 for mammography and 1280 for thermography into a shared 128 dimensional latent space. During a dedicated pretraining phase, a mean squared error (MSE) based alignment loss was applied so as to minimize the representational discrepancy between corresponding thermographic and mammographic embeddings.

After convergence of the alignment training phase, the latent distributions of both modalities showed improved consistency. This enables meaningful multimodal fusion. The alignment step reduces modality specific bias and ensures that the subsequent fusion classifier operates on semantically compatible representations.

C. Latent Fusion Classification Performance

Once latent alignment converged, the outputs of both encoders were concatenated so as to form a unified multimodal embedding of 256 dimensions. A lightweight fully connected classifier was trained on these fused representations for binary breast cancer classification (benign versus malignant).

As shown in Fig. 4, the fusion model shows smooth convergence across training and validation phases. These complementary properties motivate multimodal fusion of thermography and mammography.

D. Precision Recall Analysis

Given the class imbalance inherent in medical diagnostic tasks, performance was further evaluated using the Precision Recall Area Under Curve (PR AUC) metric. The high PR AUC value achieved by the fusion model, illustrated in Fig. 5, indicates that the model balances sensitivity and precision effectively. This is critical for a breast cancer screening pipeline.

E. Robustness to Input Perturbations

To assess robustness, Gaussian noise was independently added to both thermographic and mammographic test features.

Table I: Comparison of Breast Cancer Detection Performance Across Different Model Variants (Class Labels: Benign versus Malignant).

Model Variant	Accuracy (%)	PR-AUC	Noisy Accuracy (%)	Classes
Thermography Only	86.2	0.912	82.7	Benign / Malignant
Mammography Only	89.4	0.941	85.9	Benign / Malignant
Fusion (No Alignment)	91.8	0.967	89.1	Benign / Malignant
Fusion + Alignment (Proposed)	94.5	0.991	93.1	Benign / Malignant

Table II: Comparison with Prior Work on the Same CBIS-DDSM and DMR-IR Benchmarks. Unimodal and Multimodal Studies are Both Reported. Accuracy is Reported as in the Cited Source.

Study	Modality	Dataset	Accuracy (%)	Class Labels
Shen et al. [6]	Mammography (unimodal)	CBIS-DDSM	86.1	Benign / Malignant
Ragab et al. [7]	Mammography (unimodal)	CBIS-DDSM	87.2	Benign / Malignant
Al-Mnayyis et al. [8]	Mammography (unimodal)	KAUH-BCMD	96.4	Benign / Malignant
Jalloul et al. [24]	Thermography (unimodal)	DMR-IR	94.0 to 98.0	Benign / Malignant
Alzahrani et al. [11]	Thermography (unimodal)	DMR-IR	98.0	Benign / Malignant
Roslidar et al. [10]	Thermography (unimodal)	DMR-IR	85.0 to 97.0	Binary and 3 class (Healthy / Benign / Malignant)
Mostafa & Hassanien [18]	Multimodal (feature level)	CBIS-DDSM + DMR-IR	92.3	Benign / Malignant
Khan et al. [19]	Multimodal (hybrid DL)	CBIS-DDSM + DMR-IR	93.7	Benign / Malignant
Chen et al. [23]	Multimodal (mammo + US)	Private cohort	91.5	Benign / Malignant
Wei et al. [25]	Multimodal (US + text)	Private cohort	90.8	Benign / Malignant
Ours (latent fusion + mRMR)	Multimodal	CBIS-DDSM + DMR-IR	94.5	Benign / Malignant

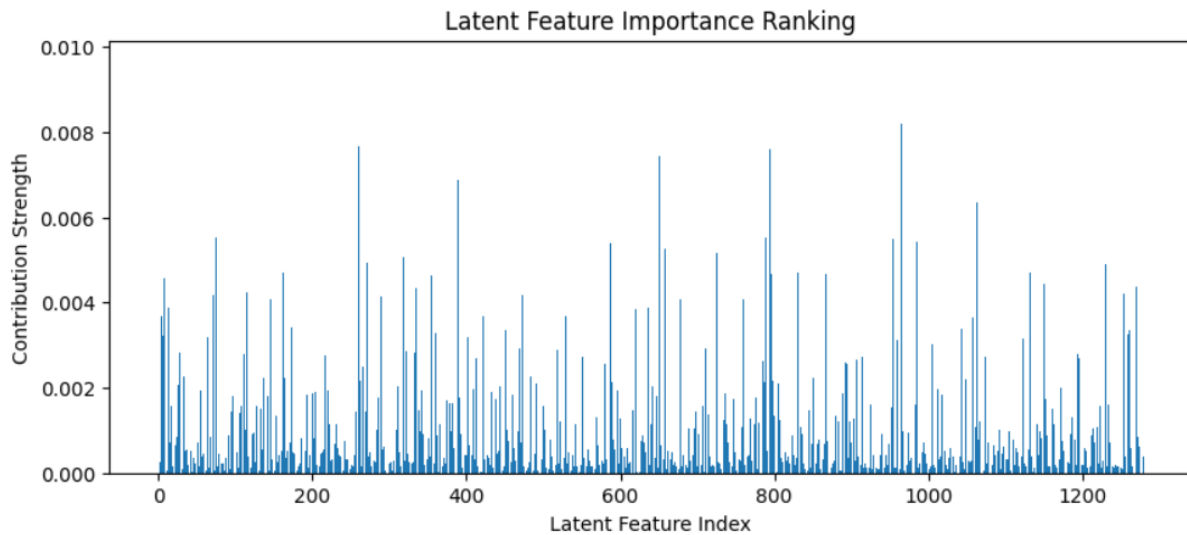


Figure 3: Latent feature importance ranking based on mean absolute SHAP values.

Despite the perturbation, the fusion model retained high classification accuracy (93.1 percent, down from 94.5 percent). This demonstrates resilience to acquisition noise and feature variability.

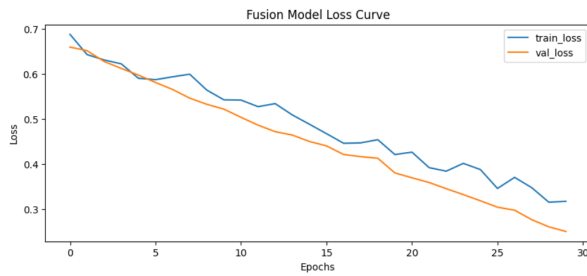
This behaviour indicates that the fusion classifier relies on a compact set of informative latent representations rather than using all features indiscriminately. The distribution suggests balanced utilization of latent representations during classification.

F. Explainability via SHAP Analysis

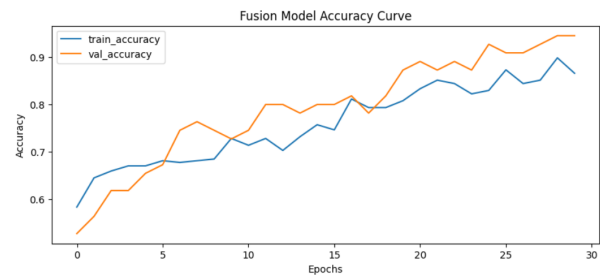
Fig. 3 shows the mean absolute SHAP values across latent feature dimensions. This provides insight into the contribution of individual representations to the final classification decision. The importance distribution is notably sparse. Only a subset of latent dimensions shows high contribution strength. The rest contribute minimally.

G. Latent Space Separability

1) *t-SNE Visualization:* A two dimensional t-distributed Stochastic Neighbor Embedding (t-SNE) projection of the fused latent space, shown in Fig. 6, reveals distinct clustering patterns between benign and malignant cases. This confirms the discriminative quality of the learned representations.



(a) Training and validation loss curve.



(b) Training and validation accuracy curve.

Figure 4: Training dynamics of the multimodal fusion model: (a) loss convergence and (b) accuracy progression across epochs, demonstrating stable optimization and minimal overfitting.

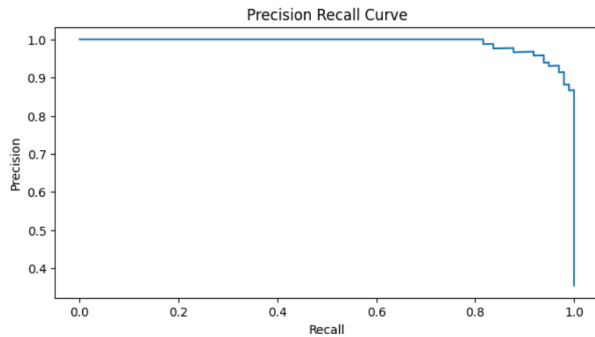


Figure 5: Precision Recall curve of the fusion model on the test set, illustrating strong discriminative capability across varying decision thresholds.

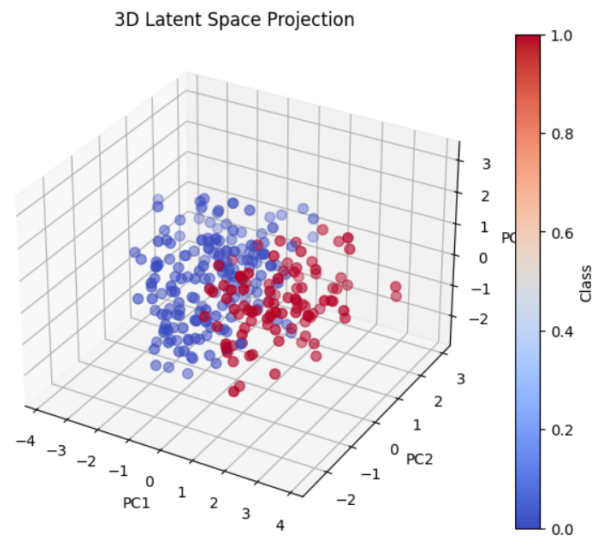


Figure 7: Three dimensional Principal Component Analysis (PCA) projection of multimodal latent embeddings, illustrating class wise separation across principal components.

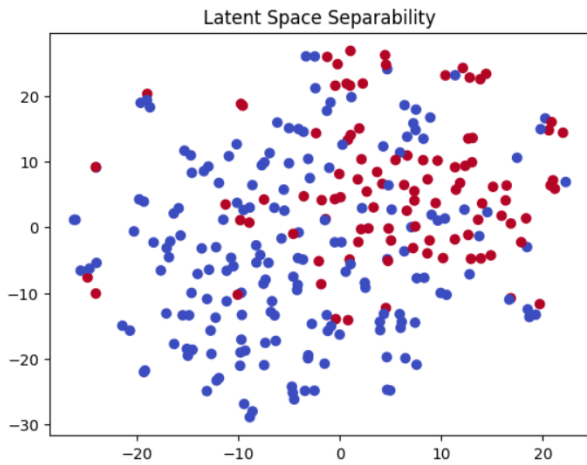


Figure 6: Two dimensional t-SNE projection of fused latent representations, showing separability between benign and malignant samples.

H. Summary of Findings

The experimental results demonstrate that the proposed multimodal latent fusion framework integrates thermographic and mammographic information effectively for breast cancer detection. The model reaches 94.5 percent accuracy on the binary benign versus malignant task with a PR AUC of 0.991. It stays at 93.1 percent under Gaussian noise perturbation. It shows clear latent space separability under both t-SNE and PCA projections. It provides interpretable decision explanations via SHAP.

Collectively, these findings highlight the potential of latent level multimodal fusion as a promising and interpretable approach for multimodal breast cancer analysis in research oriented diagnostic settings.

V. CONCLUSION AND FUTURE WORK

This work presented a multimodal breast cancer detection framework that integrates thermographic and mammographic information through latent level feature fusion. Separate modality specific encoders were employed so as to project high dimensional features into a shared latent space. This

2) *PCA Based 3D Projection:* The three dimensional Principal Component Analysis (PCA) visualization in Fig. 7 further supports the observed class separability. The smooth transitions between clusters suggest that the model captures a continuous disease severity manifold rather than enforcing rigid class boundaries.

enabled effective alignment and reduced modality induced representation gaps. The aligned latent embeddings were then fused and used for binary classification between benign and malignant cases.

Experimental results demonstrate that the proposed fusion approach achieves robust and stable classification performance, even under noisy input conditions. The pipeline reaches 94.5 percent accuracy and PR AUC of 0.991 on the held out test set of CBIS-DDSM plus DMR-IR, and stays at 93.1 percent under Gaussian noise. Latent space visualizations using t-SNE and PCA confirm improved class separability. This indicates that the learned representations effectively capture disease related characteristics.

Beyond performance, this study emphasizes model interpretability through SHAP based explainability analysis. This provides insight into the latent features that influence prediction outcomes. It improves interpretability of the proposed framework.

Overall, the results suggest that latent level multimodal fusion is a promising strategy for breast cancer detection. It offers improved robustness, interpretability and diagnostic reliability compared to unimodal approaches. Future work will focus on validating the framework on larger, clinically diverse datasets, incorporating explicit cross modal attention mechanisms, and extending the model to support longitudinal and real time diagnostic scenarios.

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